

HARYANA LEET 2026 — FORMULA & QUICK REVISION SHEET

Section B: Electronics Stream Courses | Questions 26 – 50 | 25 Marks

90 Questions · 90 Minutes · +1 / No Negative Marking · English Medium · Online MCQ

b-1 Electrical Engg. (9Q)

b-2 Electronics Engg. (8Q)

b-3 Computer Engg. (8Q)

[b-1] ELEMENTS OF ELECTRICAL ENGINEERING

DC Circuit Laws

- Ohm's Law: $V = IR$ | $P = VI = I^2R = V^2/R$
- KCL: $\sum I_{in} = \sum I_{out}$ (at a node)
- KVL: $\sum V = 0$ (around a closed loop)
- Series R: $R_{eq} = R_1 + R_2 + \dots$ | Parallel: $1/R_{eq} = 1/R_1 + 1/R_2 + \dots$
- Thevenin: Replace with V_{th} (open-circuit V) in series with R_{th}
- Norton: Replace with $I_N = V_{th}/R_{th}$ in parallel with $R_N = R_{th}$
- Superposition: each source acts alone; sum algebraically
- Max Power Transfer: $R_L = R_{th} \rightarrow P_{max} = V_{th}^2/4R_{th}$

Magnetic Circuits & Faraday's Laws

- Faraday 1st: EMF is induced when magnetic flux changes
- Faraday 2nd: $e = -N \cdot d\Phi/dt$ (induced EMF)
- Lenz's Law: induced current opposes the change causing it
- Reluctance $S = l/(\mu A)$ (analogous to resistance in mag. circuit)

AC Circuits

- $V_{rms} = V_m/\sqrt{2} \approx 0.707 V_m$
- $X_L = 2\pi fL$ (inductive reactance, Ω)
- $X_C = 1/(2\pi fC)$ (capacitive reactance, Ω)
- $Z = \sqrt{R^2 + (X_L - X_C)^2}$ (impedance, Ω)
- Power factor $\cos\phi = R/Z$ | Real power $P = VI\cos\phi$ (W)
- Reactive power $Q = VI\sin\phi$ (VAR) | Apparent $S = VI$ (VA)
- Resonance: $f_r = 1/(2\pi\sqrt{LC})$, $X_L = X_C$, $Z = R$ (min), $I = \max$

Transformers

- Turns ratio: $V_1/V_2 = N_1/N_2 = I_2/I_1$
- Efficiency $\eta = P_{out}/P_{in} \times 100\%$ (ideal $\eta=100\%$)
- Step-up: $N_2 > N_1$, $V_2 > V_1$ | Step-down: $N_2 < N_1$
- Core losses: Hysteresis + Eddy current (reduced by lamination)

DC Machines

- DC Generator: mechanical energy $\rightarrow$ electrical; works on Faraday's law
- Generated EMF: $E = (P\Phi NZ)/(60A)$ (P =poles, Z =conductors, A =paths)
- Motor: Back EMF $E_b = V - I_a R_a$
- Speed: $N \propto E_b/\Phi$ | Torque $T \propto \Phi I_a$

3-Phase Induction Motor

- Synchronous speed: $N_s = 120f/P$ (rpm)
- Slip: $s = (N_s - N)/N_s$ | At start $s=1$, Full load $s \approx 0.04-0.06$
- Rotor freq: $f_r = s \cdot f$
- Squirrel cage: robust, simple | Slip ring: external resistance control

Transmission & Distribution

- High voltage $\rightarrow$ low $I \rightarrow$ low I^2R loss $\rightarrow$ economical transmission
- 3-phase 3-wire: most economical | Power $= \sqrt{3} \cdot V_L \cdot I_L \cdot \cos\phi$

[b-2] ELEMENTS OF ELECTRONICS ENGINEERING

Semiconductor Diodes

- PN junction: forward bias → conducts; reverse bias → blocks
- **Barrier potential:** $Si \approx 0.7V$, $Ge \approx 0.3V$
- Zener diode: voltage regulator in reverse breakdown region
- LED: emits photons in forward bias | Photodiode: reverse biased, light → current
- **Half-wave rectifier:** $V_{dc} = V_m/\pi$ | **Full-wave:** $V_{dc} = 2V_m/\pi$

BJT Transistor

- $I_E = I_B + I_C$
- $\beta (hFE) = I_C/I_B$ | $\alpha = I_C/I_E$ | $\beta = \alpha/(1-\alpha)$
- CE: current+voltage gain, phase inversion; most used config
- CB: voltage gain only | CC (Emitter follower): current gain, $V_{gain} \approx 1$
- Regions: Active (amplifier), Saturation (switch ON), Cut-off (switch OFF)

FET & MOSFET

- FET: voltage controlled device; I_D controlled by V_{GS}
- JFET: depletion mode only | MOSFET: depletion or enhancement mode
- **Transconductance** $g_m = \Delta I_D / \Delta V_{GS}$

Op-Amp

- **Inverting amplifier:** $A_v = -R_f/R_{in}$
- **Non-inverting:** $A_v = 1 + R_f/R_1$
- **Voltage follower (buffer):** $A_v = 1$
- Ideal op-amp: $A_v \rightarrow \infty$, $R_{in} \rightarrow \infty$, $R_{out} \rightarrow 0$, $BW \rightarrow \infty$
- Virtual ground: inverting input $\approx 0V$ when negative feedback used

Digital Electronics

- **Binary → Decimal:** $\Sigma(\text{bit} \times 2^{\text{position}})$ e.g. $1011=11$
- **Decimal → Binary:** repeated division by 2, read remainders upward
- **Hex:** 0-9, A-F | 1 hex digit = 4 binary bits
- Basic gates: AND($\cdot$), OR($+$), NOT($'$). NAND & NOR are universal gates
- **De Morgan:** $(A \cdot B)' = A' + B'$ | $(A + B)' = A' \cdot B'$
- Boolean: $A+A=A$, $A \cdot A=A$, $A+A'=1$, $A \cdot A'=0$, $A+1=1$, $A \cdot 0=0$
- SOP: Sum of Products (OR of ANDs) | POS: Product of Sums
- FF types: SR (set-reset), D (data), JK (universal), T (toggle)
- Registers: store data | Counters: ripple, synchronous

Modulation & Communication

- AM: amplitude of carrier varies with signal; $BW=2f_m$
- FM: frequency varies; $BW=2(\Delta f + f_m)$; less noise than AM
- **Modulation index:** AM: $m_a = A_m/A_c$ | FM: $m_f = \Delta f/f_m$

Industrial Electronics (SCR, DIAC, TRIAC)

- SCR: turns ON with gate pulse; turns OFF when I falls below I_h (holding current)
- DIAC: bidirectional, no gate; breaks over at V_{BO} in both directions
- TRIAC: bidirectional SCR with gate; used in AC power control
- ✓ **Applications:** motor speed control, light dimmers, heater control, UPS

Measurements & Instruments

- Ammeter: very low R , connected in series with circuit
- Voltmeter: very high R , connected in parallel with load
- Wattmeter: measures real power; two-wattmeter method for 3-phase
- CRO: displays waveforms; measures V , I , f , phase, T
- **Frequency from CRO:** $f = 1/T$ (T = time period read from screen)
- Wheatstone bridge: $R_x = R_2 R_3 / R_1$ (null condition)
- Transducers: LVDT (displacement), RTD/thermocouple (temperature)

[b-3] ELEMENTS OF COMPUTER ENGINEERING

Computer Organisation

- CPU: ALU + CU + Registers | Memory: Primary (RAM/ROM) + Secondary
- Instruction cycle: Fetch → Decode → Execute → Store

- RISC: few simple instructions | CISC: many complex instructions
- Hard-wired CU: fast, fixed | Microprogrammed CU: flexible

Memory System

■ **1 bit < 1 nibble(4) < 1 byte(8) < 1 KB(1024B) < 1 MB(1024KB) < 1 GB**

- Memory hierarchy (speed↓, cost↓): Reg→Cache→RAM→HDD/SSD
- Cache: L1(fastest,smallest), L2, L3 — reduces avg memory access time
- Virtual memory: uses disk space as extended RAM via paging/segmentation
- ROM: non-volatile; RAM: volatile; SRAM: fast cache; DRAM: main memory

Operating System

- Process states: New→Ready→Running→Waiting→Terminated
- Scheduling: FCFS, SJF (optimal avg waiting time), Round Robin (time quantum), Priority
- Deadlock conditions: Mutual Exclusion, Hold & Wait, No Preemption, Circular Wait
- Critical section: mutual exclusion via semaphores / mutex locks
- Paging: fixed-size pages; Segmentation: variable-size logical units

C Programming

- Data types: int(2/4B), float(4B), double(8B), char(1B)
- Operators: arithmetic(+, -, *, /, %), relational(<, >, =), logical(&, ||, !)
- **Pointer: int *p; p=&x; *p dereferences; array arr[i]=*(arr+i)**
- Functions: call by value (copy) | call by reference (pointer passed)
- Struct: groups different types | Union: shares memory among members
- File I/O: fopen(mode), fclose, fprintf, fscanf, fread, fwrite

✓ **Algorithm → Flowchart → Code → Compile → Test → Debug**

Data Structures & Languages

- Compiler: translates whole program at once | Interpreter: line by line
- Stack: LIFO | Queue: FIFO | Linked list: dynamic memory
- Sorting: Bubble $O(n^2)$, Quick $O(n \log n)$ avg, Merge $O(n \log n)$

Computer Networking

- LAN (<1km), MAN (city), WAN (global) | Internet: global WAN
- Topology: Bus, Ring, Star, Mesh
- **IP Address: 32-bit (IPv4) e.g. 192.168.1.1 | IPv6: 128-bit**
- OSI 7 layers: Physical, Data Link, Network, Transport, Session, Presentation, Application
- TCP: reliable, connection-oriented | UDP: fast, connectionless
- HTTP(80), HTTPS(443), FTP(21), SMTP(25), DNS(53), DHCP(67)